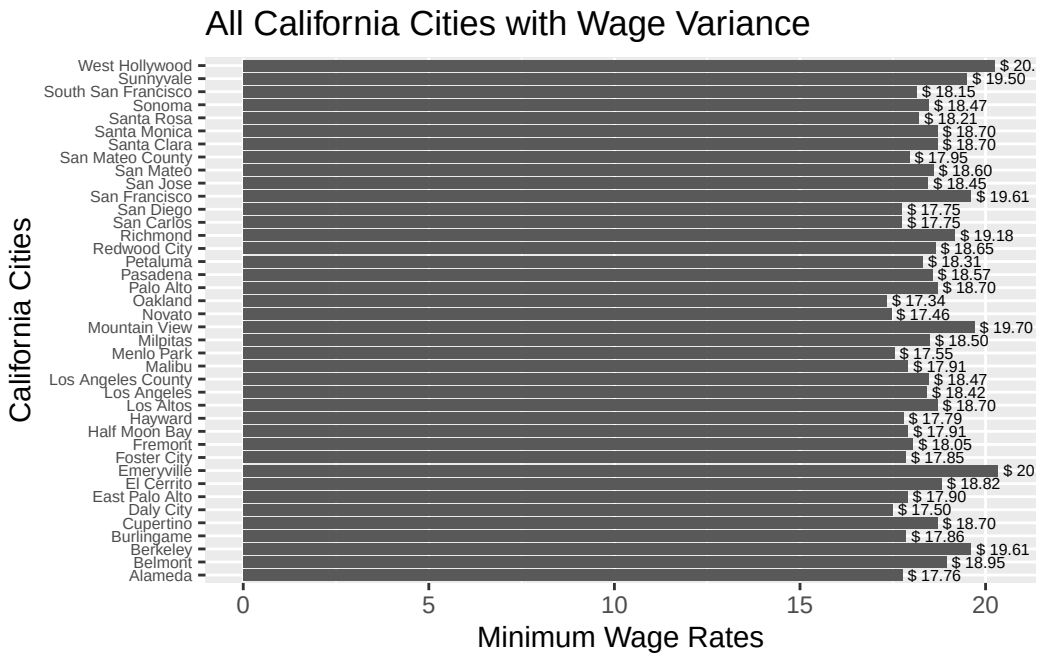
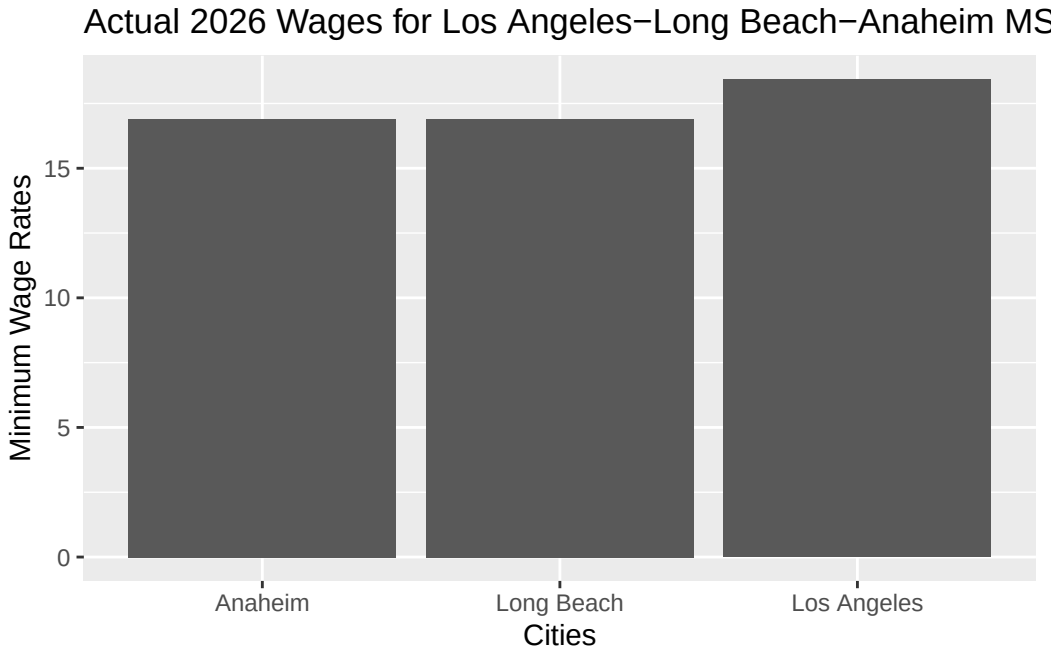
project draft

```
# load in packages
library(tidyverse)
library(readr)
library(readxl)

## Index of ALL California Cities/Counties with wage variance

# read in CSV file
cali_wages <- read_csv("CaliforniaCityMinimumWages.csv")
ggplot(cali_wages, aes(x = Rate, y = Locality)) +
  geom_col() +
  geom_text(aes(label = paste("$", format(round(Rate, 2)))),
            hjust = -0.1,
            size = 2) +
  labs(
    y = "California Cities",
    x = "Minimum Wage Rates",
    title = "All California Cities with Wage Variance"
  ) +
  theme(
    axis.text.y = element_text(size = 6)
  )
```

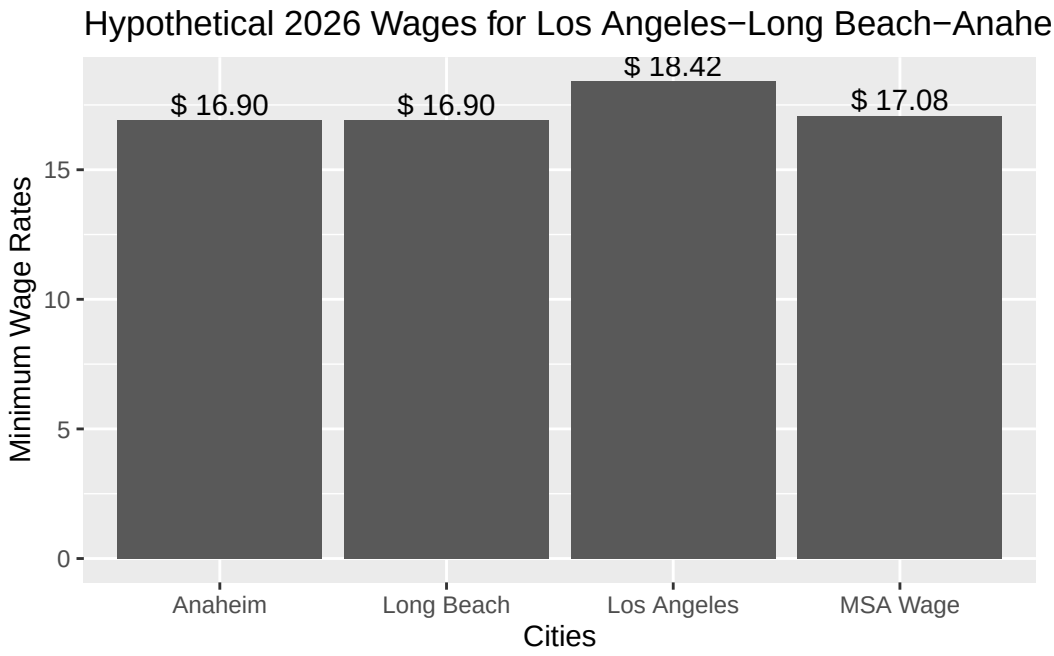



```
## FOR Los Angeles-Long Beach-Anaheim MSA!!

# create data set with LA MSA wages
og_df <- data.frame(
  cities = rep(c("Los Angeles", "Long Beach", "Anaheim")),
  wages = c(18.42, 16.90, 16.90)
)
# plot original wages
ggplot(og_df, aes(x = cities, y = wages)) +
  geom_col() +
  labs(
    x = "Cities",
    y = "Minimum Wage Rates",
    title = "Actual 2026 Wages for Los Angeles-Long Beach-Anaheim MSA"
  )
```

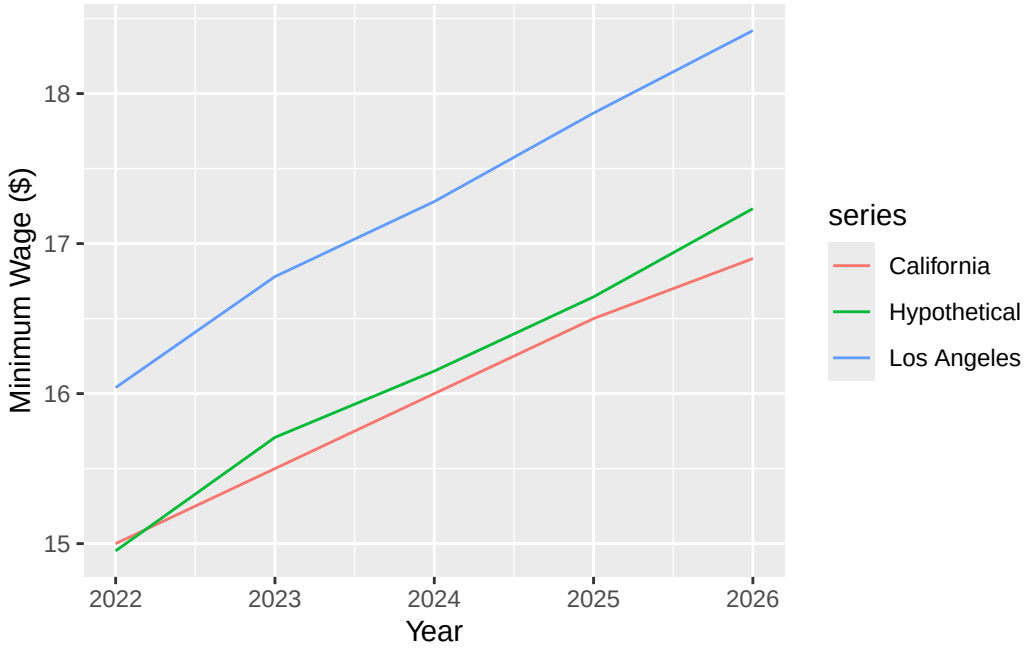



```
# read in excel file
LA_MSA <- read_excel("LA_MSA.xlsx")
# remove unnecessary rows
newLA_df <- LA_MSA[-c(1:10), ] # now df displays CPI-W by month and year

# solve for the rate of change
new_avg <- mean(c(as.numeric(newLA_df[12, 2:7]),
                  as.numeric(newLA_df[11, 8:13])), na.rm = TRUE)
old_avg <- mean(c(as.numeric(newLA_df[11, 2:7]),
                  as.numeric(newLA_df[10, 8:13])), na.rm = TRUE)
x <- new_avg/old_avg
# finding the average for 2026, then diving by the average for 2025

# calculate new MSA wage
LAnew_wage <- (16.50 * x)

# create data set which combines actual wages with the new MSA wage
compare_LAdf <- data.frame(
  cities = c("Los Angeles", "Long Beach", "Anaheim", "MSA Wage"),
  wages = c(18.42, 16.90, 16.90, LAnew_wage)
)

# plot new and old wages
```

```r
ggplot(compare_LAdf, aes(x = cities, y = wages)) +
  geom_col() +
  geom_text(aes(label = paste("$", format(round(wages, 2)))),
            vjust = -0.25) +
  labs(
    x = "Cities",
    y = "Minimum Wage Rates",
    title = "Hypothetical 2026 Wages for Los Angeles-Long Beach-Anaheim MSA"
  )
```




```r
# make year column as.numeric
newLA_df[[1]] <- as.numeric(newLA_df[[1]], na.rm = TRUE)

# create a function to calculate rates of changes for the past 5 years
avg_function <- function(year) {
  year_col <- newLA_df[[1]]
  prev <- newLA_df[year_col == year - 1, 8:13]
  now <- newLA_df[year_col == year, 2:7]
  mean(as.numeric(c(unlist(prev), unlist(now))), na.rm = TRUE)
}
years <- 2022:2026 # the past 5 years
```

```r
# create a data frame using the rates from above
rates <- data.frame(
  year = years,
  new_avg = sapply(years, avg_function),
  old_avg = sapply(years - 1, avg_function)
)
rates$x <- rates$new_avg / rates$old_avg # calculate rates of change

# calculate hypothetical minimum wages for the LA MSA area from 2022-2026
year_one <- rates$x[1] * 14.00 # $14.00 was California's statewide minimum wage in 2021
year_two <- rates$x[2] * year_one # 2023
year_three <- rates$x[3] * year_two # 2024
year_four <- rates$x[4] * year_three # 2025
year_five <- rates$x[5] * year_four # 2026

# create data frames for the hypothetical, California actual, and LA actual
hypo_wagedf <- data.frame(
  year = rates$year,
  wage = c(year_one, year_two, year_three, year_four, year_five)
) # hypothetical

cali_wagedf <- data.frame(
  year = rates$year,
  wage = c("15.00", "15.50", "16.00", "16.50", "16.90")
) # california

LA_wagedf <- data.frame(
  year = rates$year,
  wage = c("16.04", "16.78", "17.28", "17.87", "18.42")
) # LA

# rename to create separate columns
hypo_wagedf$series <- "Hypothetical"
cali_wagedf$series <- "California"
LA_wagedf$series <- "Los Angeles"

# collapse into one data frame
all_wages <- rbind(hypo_wagedf, cali_wagedf, LA_wagedf)
all_wages$wage <- as.numeric(all_wages$wage) # convert wages from strings into numbers

# line plot for all wages!
ggplot(all_wages, aes(x = year, y = wage, color = series)) +
```

```
  geom_line() +
  labs(
    x = "Year",
    y = "Minimum Wage ($)",
  )
```